

Statistical Inference & Sampling — Solutions

Full Step-by-Step | Trap Alerts | Key Insights | Fast Methods | Desmos Verdicts

Q1

Easy

Answer: C) Random sample from colleges across the US

Step-by-Step Solution:

To generalize to all US college students, the sample must be representative of that population.

A random sample drawn from colleges across the entire US provides this representativeness.

Options A, B, and D all sample from single institutions, limiting generalizability.

Why Other Options Are Wrong:

A) Only one university — not representative of all US colleges.

B) Only one community college — different demographics than all US students.

D) One class at one school — extreme convenience sample.

TRAP ALERT

Large sample size (option A has 500) does not fix non-representativeness. A biased sample of 10,000 is worse than a random sample of 500.

KEY INSIGHT

Generalizability requires **RANDOM SAMPLING** from the target population. Size helps precision, but randomness ensures representativeness.

FAST METHOD

Random sample from the target population = generalizable results. Period.

VERDICT

Hand-solve. Conceptual — 5 seconds.

Q2

Easy

Answer: B) 250

Step-by-Step Solution:

Capture-recapture formula: $N = (\text{tagged} \times \text{second capture}) / \text{recaptured}$.

$N = (40 \times 50) / 8 = 2000 / 8 = 250$.

Why Other Options Are Wrong:

A) 200 — arithmetic error.

C) 320 — uses wrong formula.

D) 400 — multiplies $40 \times 50 / 5$ or similar error.

TRAP ALERT

Students sometimes confuse which number goes where. The formula assumes that the proportion of tagged fish in the second capture equals the proportion of tagged fish in the whole lake.

KEY INSIGHT

$\text{tagged}/\text{total} = \text{recaptured}/\text{second capture}$. Solve for total: $\text{total} = \text{tagged} \times \text{second capture} / \text{recaptured}$.

FAST METHOD

$$N = 40 \times 50 / 8 = 250.$$

VERDICT

Hand-solve. One multiplication and one division — 10 seconds.

Q3

Easy

Answer: C) Plausible that true proportion is between 50% and 58%

Step-by-Step Solution:

Margin of error creates a confidence interval: $54\% \pm 4\% = [50\%, 58\%]$.

This means it is plausible that the true population proportion falls in this range.

We cannot say "exactly 54%" because there is sampling variability.

Why Other Options Are Wrong:

- A) Surveys provide estimates, not exact values.
- B) This describes the sample, not the population — and the interval is about the population.
- D) Survey results do not predict election outcomes with certainty.

TRAP ALERT

Option B refers to the "surveyed voters" — the margin of error applies to the POPULATION estimate, not to the sample itself. The sample result is exactly 54%.

KEY INSIGHT

Margin of error creates a confidence interval for the POPULATION parameter, not the sample statistic.

FAST METHOD

$54\% \pm 4\% = [50\%, 58\%]$. This is a plausible range for the true population proportion.

VERDICT

Hand-solve. Interpretation — 10 seconds.

Q4

Easy

Answer: C) No, sample is not representative

Step-by-Step Solution:

The 50 newest customers are not representative of ALL customers.

New customers may have different experiences than long-term customers.

Generalizing from a non-representative sample is not valid.

Why Other Options Are Wrong:

- A) Sample size does not fix non-representativeness.
- B) The high rate does not make the sample representative.
- D) The issue is representativeness, not margin of error.

TRAP ALERT

A convenience sample (newest customers) is biased regardless of size. Only random sampling from all customers allows generalization.

KEY INSIGHT

Sample must be representative of the population to make valid inferences. Convenience samples are biased.

FAST METHOD

Newest customers are not equal to all customers. Cannot generalize.

VERDICT

Hand-solve. Conceptual — 5 seconds.

Q5

Easy

Answer: B) News website click-to-participate poll

Step-by-Step Solution:

Voluntary response: participants choose whether to participate.

A website poll where visitors click to participate is voluntary — people with strong opinions are more likely to respond.

Why Other Options Are Wrong:

- A) Random selection from a list is a random sample, not voluntary.
- C) Random number generator = random sample.
- D) Systematic sampling (every 5th person) is not voluntary.

TRAP ALERT

Voluntary response samples are biased because people with extreme opinions (positive or negative) are more likely to participate.

KEY INSIGHT

Voluntary response = self-selected = biased. Random selection = unbiased (when done properly).

FAST METHOD

Who chooses to participate? If the participant decides, it is voluntary response.

VERDICT

Hand-solve. Definition recall — 5 seconds.

Q6

Easy

Answer: B) Association, not causation

Step-by-Step Solution:

The study is observational (no random assignment).

Observational studies can show association but not causation.

There may be confounding variables (e.g., students who eat breakfast may have better study habits, more sleep, etc.).

Why Other Options Are Wrong:

- A) Cannot claim causation from an observational study.
- C) This is a causal recommendation not supported by the study design.
- D) This reverses the direction and claims causation.

TRAP ALERT

The DSAT repeatedly tests this: observational study = association only. Experiment with random assignment = causation possible.

KEY INSIGHT

Only randomized experiments can establish causation. Observational studies establish association/correlation.

FAST METHOD

No random assignment = observational = association only.

VERDICT

Hand-solve. Key distinction — 5 seconds.

Q7

Medium

Answer: B) Between 45% and 51%, race is too close to call

Step-by-Step Solution:

48% $\pm$ 3% gives a confidence interval of [45%, 51%].

Since 50% (the threshold to win) falls within this interval, the race is too close to call.

We cannot determine with confidence whether Candidate A has majority support.

Why Other Options Are Wrong:

A) The interval includes values above 50%, so A might still win.

C) 48% is the sample estimate, not the exact population value.

D) This misinterprets confidence level — 95% refers to the method, not individual polls.

TRAP ALERT

When the confidence interval straddles the 50% threshold, neither candidate can be declared the likely winner. This is a "too close to call" scenario.

KEY INSIGHT

If the confidence interval includes the threshold value (like 50%), the result is not conclusive.

FAST METHOD

45% to 51% includes 50%. Too close to call.

VERDICT

Hand-solve. Interval analysis — 10 seconds.

Q8

Medium

Answer: B) Medication caused lower blood pressure

Step-by-Step Solution:

The study used random assignment (participants randomly assigned to two groups).

Random assignment controls for confounding variables.

With random assignment and a significant result, we can conclude causation.

The medication caused the observed difference in blood pressure.

Why Other Options Are Wrong:

- A) This is weaker than what the study supports — random assignment allows causal claims, not just association.
- C) Reverses cause and effect.
- D) We cannot conclude the placebo had "no effect" — we know the medication group did better.

TRAP ALERT

Random assignment is the key. Many students default to "association only" for all studies, but randomized experiments CAN establish causation.

KEY INSIGHT

Randomized experiment = can claim causation. Observational study = association only. This is the #1 rule for DSAT inference questions.

FAST METHOD

Random assignment? Yes -> causation is valid.

VERDICT

Hand-solve. Check for random assignment — 5 seconds.

Q9

Medium

Answer: B) Difference could be due to sampling variability

Step-by-Step Solution:

National: 62% +/- 2% gives a confidence interval of [60%, 64%].

State X: 58%. The state survey of 400 adults has its own margin of error (approximately $1/\sqrt{400} = 5\%$), giving a CI of roughly [53%, 63%].

The state CI [53%, 63%] overlaps with the national CI [60%, 64%].

When confidence intervals overlap, the difference between the two estimates could be due to normal sampling variability.

We cannot conclusively say State X differs from the national average.

Why Other Options Are Wrong:

A) Cannot conclude "significantly less" without proper hypothesis testing and considering both margins of error.

C) A survey of 400 is not unreliable — it just has a larger margin of error.

D) They do not necessarily contradict each other — overlapping intervals mean they could reflect the same underlying value.

TRAP ALERT

Students see 58% vs 62% and conclude they are "significantly different." But both surveys have margins of error, and when you account for both, the intervals overlap.

KEY INSIGHT

When comparing two survey results, consider both margins of error. If the intervals overlap, the difference may not be statistically significant.

FAST METHOD

Both surveys have uncertainty. The state estimate (58%) is within plausible range when you factor in both margins. Likely sampling variability.

VERDICT

Hand-solve. Compare intervals — 10 seconds.

Q10

Medium

Answer: 400

Step-by-Step Solution:

Capture-recapture: $N = (\text{tagged} \times \text{second capture}) / \text{recaptured}$.

$$N = (60 \times 80) / 12 = 4800 / 12 = 400.$$

TRAP ALERT

Grid-in: students might divide in the wrong order ($12 \times 80 / 60 = 16$, or $60 \times 12 / 80 = 9$). Set up the proportion carefully: $\text{tagged}/N = \text{recaptured}/\text{second capture}$.

KEY INSIGHT

Proportion: $60/N = 12/80$. Cross-multiply: $60 \times 80 = 12N$. $N = 4800/12 = 400$.

FAST METHOD

$$N = 60 \times 80 / 12 = 400.$$

VERDICT

Hand-solve. One proportion — 10 seconds.

Q11

Medium

Answer: C) Randomly assign patients

Step-by-Step Solution:

Only randomized experiments can establish causation.

Random assignment to treatment groups controls for confounding variables.

This is the gold standard for causal inference.

Why Other Options Are Wrong:

A) Observational — patients self-select, introducing confounding variables.

B) Surveys measure opinions, not causal effects.

D) Comparing hospitals is observational — hospital differences introduce confounders.

TRAP ALERT

Option A looks scientific but is observational (patients chose the technique). Only RANDOM assignment eliminates confounders.

KEY INSIGHT

Causal relationship requires: (1) random assignment, (2) control group, (3) treatment group.

FAST METHOD

Random assignment = causal. Self-selection = observational.

VERDICT

Hand-solve. Identify study design — 5 seconds.

Q12

Medium

Answer: B) 95% confident true average is between \$134 and \$150**Step-by-Step Solution:**

Confidence interval: $\$142 \pm \$8 = [\$134, \$150]$.

Interpretation: We are 95% confident that the true population mean falls in this interval.

This is about the POPULATION PARAMETER, not individual values.

Why Other Options Are Wrong:

- A) The CI is about the mean, not about individual household bills.
- C) The CI is not about monthly variation over time.
- D) Same error as A — CI is not about individual values.

TRAP ALERT

The confidence interval estimates the POPULATION MEAN, not individual values. "95% of households pay..." is wrong because the CI is about the average, not the distribution of individual bills.

KEY INSIGHT

Confidence intervals estimate population parameters (like the mean), not individual data points. This is a critical distinction.

FAST METHOD

CI is about the population mean. "95% confident the true average is between..."

VERDICT

Hand-solve. Interpretation — 5 seconds.

Q13

Medium

Answer: C) Nonresponse bias**Step-by-Step Solution:**

Only 30% of surveys were returned — 70% did not respond.

Parents who returned surveys may have stronger opinions (for or against).

This creates nonresponse bias: respondents systematically differ from non-respondents.

Why Other Options Are Wrong:

- A) With 30% of surveys returned, sample size could be adequate.
- B) The question does not mention biased wording.
- D) Margin of error is a concern, but nonresponse bias is more fundamental.

TRAP ALERT

Low response rate (30%) is a red flag for nonresponse bias. Even with a large initial mailing, if only certain types of people respond, the results are biased.

KEY INSIGHT

Nonresponse bias occurs when people who respond differ systematically from those who do not. Low response rates amplify this concern.

FAST METHOD

70% did not respond. Are non-responders different from responders? Probably yes -> nonresponse bias.

VERDICT

Hand-solve. Identify bias type — 10 seconds.

Q14

Medium

Answer: B) Study B, random assignment controls confounders

Step-by-Step Solution:

Study A is observational (large sample but no random assignment).

Study B is a randomized experiment (smaller sample but random assignment).

Random assignment is the key to establishing causation.

Study B provides stronger evidence of a CAUSAL relationship, despite the smaller size.

Why Other Options Are Wrong:

A) Sample size helps with precision but does not establish causation.

C) Random assignment (Study B) is fundamentally stronger for causal claims.

D) Randomized experiments can provide causal evidence.

TRAP ALERT

Students often think bigger sample = better study. For CAUSAL claims, study design (random assignment) matters more than sample size.

KEY INSIGHT

A small randomized experiment beats a large observational study for establishing causation. Size helps precision; design determines what you can conclude.

FAST METHOD

Random assignment > large sample size for causal claims.

VERDICT

Hand-solve. Study design hierarchy — 5 seconds.

Q15

Medium

Answer: 0.03

Step-by-Step Solution:

Margin of error = $1/\sqrt{n} = 1/\sqrt{900} = 1/30 = 0.0333\dots$

Rounded to nearest hundredth: 0.03.

TRAP ALERT

Grid-in: students might confuse $\sqrt{900} = 30$ (not 300 or 90). Also, $1/30 = 0.0333\dots$, which rounds to 0.03, not 0.04.

KEY INSIGHT

The margin of error formula $1/\sqrt{n}$ is a simplified version. As n increases, the margin of error decreases — but at a square-root rate, not linearly.

FAST METHOD

$$\sqrt{900} = 30. \ 1/30 = 0.033... = 0.03.$$

VERDICT

Hand-solve. Quick calculation — 10 seconds.

Q16

Medium

Answer: B) Difference unlikely due to chance alone

Step-by-Step Solution:

$p < 0.05$ means: if there were no real difference, the probability of observing a result this extreme is less than 5%.

This means the observed difference is unlikely to have occurred by random chance alone.

It does NOT mean the difference is large or practically meaningful.

Why Other Options Are Wrong:

- A) Statistical significance does not mean practical significance.
- C) The confidence level is not the probability of being "correct."
- D) p-value says nothing about study design quality.

TRAP ALERT

Students confuse statistical significance with practical significance. A result can be statistically significant but trivially small (especially with large samples).

KEY INSIGHT

p-value = probability of observing the data (or more extreme) IF the null hypothesis is true. Small p = evidence against the null.

FAST METHOD

$p < 0.05$ means "unlikely due to chance." That is ALL it means.

VERDICT

Hand-solve. Definition recall — 5 seconds.

Q17

Hard

Answer: A) Cut in half (approximately 2.5%)

Step-by-Step Solution:

Margin of error is proportional to $1/\sqrt{n}$.

Original: $n = 400$. New: $n = 1600$.

Ratio: $1/\sqrt{1600} / 1/\sqrt{400} = \sqrt{400}/\sqrt{1600} = 20/40 = 1/2$.

The margin of error is halved: $5\% / 2 = 2.5\%$.

Why Other Options Are Wrong:

- B) Quadrupling n halves the MOE (not quarters it).
- C) Increasing sample size always decreases MOE.
- D) Larger samples decrease variability, not increase it.

TRAP ALERT

To CUT the margin of error in half, you must QUADRUPLE the sample size (because MOE depends on $\sqrt{n}$). Students think doubling n halves MOE, but it does not.

KEY INSIGHT

MOE is proportional to $1/\sqrt{n}$. To halve MOE, quadruple n . To reduce MOE by $1/3$, multiply n by 9.

FAST METHOD

n goes from 400 to 1600 = $\times 4$. $\sqrt{4} = 2$. MOE is halved: $5/2 = 2.5\%$.

VERDICT

Hand-solve. Square root relationship — 10 seconds.

Q18

Hard

Answer: B) Caused 15% increase, unlikely due to chance

Step-by-Step Solution:

Random assignment was used $\rightarrow$ causal conclusion is valid.

The result is statistically significant $\rightarrow$ unlikely due to chance.

But we can only generalize to the experimental conditions (40 plots), not "all crops everywhere."

Why Other Options Are Wrong:

A) Overgeneralizes — cannot extend to "all crops everywhere."

C) Statistical significance says it IS meaningful (unlikely due to chance). Practical significance is separate.

D) While replication is good practice, a single well-designed experiment can support a conclusion.

TRAP ALERT

Option A is the overgeneralization trap. Experiments can establish causation, but only for the population/conditions studied, not the entire universe.

KEY INSIGHT

Random assignment $\rightarrow$ causation valid. But do not generalize beyond the experimental conditions without further evidence.

FAST METHOD

Random assignment + significant result $\rightarrow$ caused the effect IN THIS EXPERIMENT.

VERDICT

Hand-solve. Scope of inference — 10 seconds.

Q19

Hard

Answer: B) Selection bias from survey location

Step-by-Step Solution:

Surveying outside a cycling shop attracts cycling enthusiasts.

Cycling enthusiasts are more likely to support bike lanes.

This creates selection bias (also called sampling bias).

The sample is not representative of all city residents.

Why Other Options Are Wrong:

- A) 200 is a reasonable sample size — the problem is bias, not size.
- C) Margin of error is secondary to the fundamental bias issue.
- D) 82% is a valid survey result; the problem is it comes from a biased sample.

TRAP ALERT

Students might focus on sample size or margin of error, but the most significant concern is WHERE the survey was conducted. A biased sample, no matter how large, gives biased results.

KEY INSIGHT

Selection bias occurs when the sampling method systematically excludes or overrepresents certain groups. The survey location determines who is available to be sampled.

FAST METHOD

Cycling shop -> cyclists -> favor bike lanes. Selection bias.

VERDICT

Hand-solve. Identify bias type — 5 seconds.

Q20

Hard

Answer: 0.68

Step-by-Step Solution:

Proportion = number who prefer / total = $340/500 = 0.68$.

TRAP ALERT

Grid-in: some students might compute $500/340 = 1.47$ (inverted) or convert to a percentage (68%) instead of a decimal. The question asks for a decimal.

KEY INSIGHT

Proportion = part/whole. Always between 0 and 1 when expressed as a decimal.

FAST METHOD

$340/500 = 0.68$.

VERDICT

Hand-solve. Division — 5 seconds.

Q21

Hard

Answer: C) 3% probability of observing this difference if Drug X had no effect

Step-by-Step Solution:

$p = 0.03$ means: IF the drug had no real effect (null hypothesis is true), there is a 3% probability of observing a result as extreme as 30% reduction.

This is NOT the probability that the drug does not work (option A).

This is NOT the probability that the drug is effective (option B).

Why Other Options Are Wrong:

- A) The p-value is not the probability that the drug does not work.
- B) The p-value is not the probability that the drug works.
- D) The p-value says nothing about what percentage of patients respond.

TRAP ALERT

The most common p-value misinterpretation: "p = 0.03 means 3% chance the drug does not work."
This is **WRONG**. The p-value assumes the null is true and asks how likely the DATA are.

KEY INSIGHT

p-value = $P(\text{data as extreme or more} \mid \text{null is true})$. It is NOT $P(\text{null is true} \mid \text{data})$.

FAST METHOD

p = 0.03: "3% chance of this result IF the drug did nothing."

VERDICT

Hand-solve. Definition — 5 seconds.

Q22**Hard**

Answer: B) Statistically significant but may not be practically significant

Step-by-Step Solution:

$p < 0.001$ -> highly statistically significant.

But the actual effect is 0.3 months (about 9 days) of additional life.

Most people would consider 9 days of extra life to be not practically meaningful.

Large samples can detect very tiny effects that are statistically but not practically significant.

Why Other Options Are Wrong:

A) Practical significance requires a meaningful effect size, which 0.3 months is not.

C) It IS statistically significant ($p < 0.001$).

D) It IS statistically significant.

TRAP ALERT

Statistical significance (p-value) and practical significance (effect size) are separate concepts. The DSAT tests this distinction explicitly.

KEY INSIGHT

With very large samples, even tiny differences become statistically significant. Always consider both statistical and practical significance.

FAST METHOD

$p < 0.001$ (significant) + 0.3 months (tiny effect) = statistically but not practically significant.

VERDICT

Hand-solve. Concept application — 10 seconds.

Q23**Hard**

Answer: C) Working adults at their jobs

Step-by-Step Solution:

Calling landline phones between 10 AM-2 PM on weekdays.

Working adults are at their jobs during these hours and will not answer.

This creates undercoverage of working-age adults.

Why Other Options Are Wrong:

- A) Retired individuals are likely to be home during the day.
- B) Stay-at-home parents are likely to be home.
- D) Rural areas have landlines — this is not the primary issue.

TRAP ALERT

The key constraint is the TIME of the calls, not the phone type. Working adults are systematically excluded by weekday daytime calls.

KEY INSIGHT

Survey timing, method, and location all determine who can be reached. Each creates potential undercoverage of certain groups.

FAST METHOD

10 AM - 2 PM weekdays: who is NOT home? Working adults.

VERDICT

Hand-solve. Logical reasoning — 5 seconds.

Q24

Hard

Answer: No

Step-by-Step Solution:

Confidence interval: (49%, 55%).

The interval includes values below 50% (specifically, 49%).

Since the interval contains values where the candidate would lose, we cannot conclude with 95% confidence that the candidate will win.

TRAP ALERT

Grid-in (Yes/No): Students see 52% and think the candidate is ahead. But the confidence interval extends to 49%, which means losing is within the plausible range.

KEY INSIGHT

If the confidence interval for support includes values below the winning threshold, the result is not conclusive.

FAST METHOD

CI includes 49% (below 50%). Cannot conclude a win.

VERDICT

Hand-solve. Check if threshold is inside the CI — 5 seconds.

Q25

Hard

Answer: B) Confounding variable (motivation)

Step-by-Step Solution:

Students voluntarily enrolled in tutoring — this is self-selection.

Students who choose tutoring may be more motivated, organized, or concerned about grades.

These confounding variables could explain the higher scores, not the tutoring itself.

Without random assignment, we cannot separate the effect of tutoring from the effect of motivation.

Why Other Options Are Wrong:

- A) Sample size is not the issue — it is study design.
- C) Test standardization is a separate issue from confounding.
- D) p-values are not required to identify a confounding variable problem.

TRAP ALERT

Voluntary enrollment = self-selection = confounding. This is the most common critique of observational studies in education research.

KEY INSIGHT

When participants self-select into groups, any pre-existing differences between the groups (confounders) make causal claims invalid.

FAST METHOD

Volunteered for tutoring = more motivated. Motivation is a confounder.

VERDICT

Hand-solve. Identify confounding — 10 seconds.

TOPIC SUMMARY

Core Concept

Statistical inference uses sample data to make conclusions about a population. The DSAT focuses on understanding study design, sampling methods, bias, confidence intervals, and the distinction between causation and association.

The Big Two Rules

1. Random SAMPLING from a population allows GENERALIZATION to that population.
2. Random ASSIGNMENT to groups allows CAUSAL conclusions.

These are independent: a study can have one, both, or neither.

Key Vocabulary

Confidence interval: range of plausible values for the population parameter.

Margin of error: half-width of the confidence interval. Proportional to $1/\sqrt{n}$.

p-value: probability of observing data as extreme IF the null hypothesis is true.

Statistical significance: $p < \text{threshold}$ (usually 0.05).

Confounding variable: a variable that affects both the treatment and the outcome.

Types of Bias

Selection bias: sampling method systematically excludes groups.

Nonresponse bias: respondents differ from non-respondents.

Voluntary response bias: self-selected participants have extreme views.

Most Common DSAT Question Types

1. Identify the appropriate conclusion (causation vs. association).
2. Interpret a confidence interval or margin of error.
3. Identify sampling bias or limitations.
4. Capture-recapture population estimation.
5. Evaluate study design (randomized experiment vs. observational study).

Top 3 Traps to Avoid

1. Claiming causation from an observational study (no random assignment).
2. Misinterpreting confidence intervals as being about individual values instead of the population parameter.
3. Thinking large sample size fixes sampling bias (it does not).

Desmos vs. Hand-Solve Summary

95% of inference questions are conceptual and require no calculation. Desmos is only useful for capture-recapture arithmetic or margin of error calculations.